

Priyan Vaithilingam

✉ priyanmuthu@gmail.com | 🏠 priyan.info | 📧 priyanmuthu | 🌐 priyanv

Education

Harvard University

PHD IN COMPUTER SCIENCE | CGPA*: 4/4 | ADVISED BY DR. ELENA GLASSMAN

Boston, USA

July 2020 - March 2025

University of California, San Diego

M.S. IN COMPUTER SCIENCE | GPA: 3.96/4 | ADVISED BY DR. PHILIP GUO

San Diego, USA

Sept 2018 - June 2020

Indian Institute of Technology (IIT), Indore

B.TECH. IN COMPUTER SCIENCE AND ENGINEERING | GPA: 9/10

Indore, India

June 2011 - May 2015

Relevant Work Experience

Research Scientist, Apple

- Generative UI research

Seattle, USA

Apr 2025 - Present

Research Intern, Microsoft Research

- Embedding AI assistance directly into OS-level interactions like Copy-Paste.
- Created MagicCopy – to helps users smartly copy-paste data between applications using LLMs.

Redmond, USA

May 2024 - Aug 2024

Research Intern, Microsoft Research

- Re-imagining interaction techniques for AI-assisted data visualization tools
- Focusing on dynamic UI for editing and interacting with visualizations that adapts to user intent.

Redmond, USA

May 2023 - Aug 2023

Research Intern, Microsoft

- Re-imagining interaction techniques for AI-based automated programming tools
- Research on improving Intellicode user experience in Visual Studio

Remote

Sept 2021 - Aug 2022

Research Fellow, Microsoft Research India

- Honorable mention paper in CHI 2018 | Worldwide 2nd place in Microsoft Hackathon 2017.
- Research in Human Computer Interaction, Programming Languages and Information Retrieval.

Bangalore, India

Mar. 2016 - July 2018

Relevant Publications

The Way We Notice, That's What Really Matters: Instantiating UI Components with Distinguishing Variations

CHI 2026

Priyan Vaithilingam, Alan Leung, Jeffrey Nichols, Titus Barik

Semantic Commit: Helping Users Update Intent Specifications for AI Memory at Scale

UIST 2025

Priyan Vaithilingam, Munyeong Kim, Frida-Cecilia Acosta-Parenteau, Daniel Lee, Amine Mhedhbi, Elena L. Glassman, Ian Arawjo

Imagining a Future of Designing with AI: Dynamic Grounding, Constructive Negotiation, and Sustainable Motivations

DIS 2024

Priyan Vaithilingam, Ian Arawjo, Elena L. Glassman

DynaVis: Dynamically Synthesized UI Widgets for Visualization Editing

CHI 2024 (Best Paper Award)

Priyan Vaithilingam, Elena Glassman, Jeevana Priya Inala

ChainForge: A Visual Toolkit for Prompt Engineering and LLM Hypothesis Testing

CHI 2024 (Honorable Mention Award)

Ian Arawjo, Priyan Vaithilingam, Martin Wattenberg, Elena Glassman

Towards More Effective AI-Assisted Programming: A Systematic Design Exploration to Improve Visual Studio IntelliCode's User Experience

ICSE SEIP 2023

Priyan Vaithilingam, et al.

Expectation vs. Experience: Evaluating the Usability of Code Generation Tools Powered by Large Language Models

CHI LBW 2022

Priyan Vaithilingam, Tianyi Zhang, Elena L. Glassman

Assuage: Assembly Synthesis Using A Guided Exploration.

UIST 2021

Jingmei Hu, **Priyan Vaithilingam**, Stephen Chong, Margo Seltzer, Elena L. Glassman

Interpretable Program Synthesis

CHI 2021

Tianyi Zhang, Zhiyang Chen, Yuanli Zhu, **Priyan Vaithilingam**, Xinyu Wang, Elena L. Glassman

Bespoke: Interactively Synthesizing Custom GUIs from Command-Line Applications By Demonstration.

UIST 2019

Priyan Vaithilingam, Philip Guo

CodeTalk: Improving programming environment accessibility for Visually Impaired developers

CHI 2018 (**Honorable Mention Award**)

Venkatesh Potluri, **Priyan Vaithilingam**, Suresh Iyengar, Vidya Y, Manohar Swaminathan, Gopal Srinivasa